

2026-09-21 Selectivity measures from job postings

(Preliminary and subject to significant revisions)

Wang Wenzhi

September 22, 2026

Outline

① General introduction: Variable construction from job postings

The goal

The measurement problem and one potential solution

My personal thoughts

② Variables: Education/experience requirements

③ Variables: Skill requirements

What do we want to measure from job postings?

What do we want to measure from job postings?

- **The degree of selectivity: Education/experience/skill/qualification requirements.**
 - What is the hiring standard in terms of qualification? What is the standard in terms of education, degree fields, experience? What general/specific skills are required? How common/rare are them in the whole sample of job postings? Are specific license/qualifications required?
 - Why do we need this selectivity measure? **Firms using mainly the experimentation strategy may lower their hiring standard and selectivity level; while firms using mainly the screening strategy may deliberately ask for higher observable qualifications.**
- Disclosed screening procedures: Tests/interviews/other screening procedures.
- Optional: Amenities (remote work, flexibility, culture, insurance, etc).

Outline

① General introduction: Variable construction from job postings

The goal

The measurement problem and one potential solution

My personal thoughts

② Variables: Education/experience requirements

③ Variables: Skill requirements

More clarifications on the measurement problem

- Most of our selectivity indicators will be based on “ground truth.”
 - We are not measuring trust/culture/emotion from job postings – these concepts need a clear definition first.
- The main issue in this measurement problem is that we don't have any training/test data with clearly labeled and validated variables to measure education and experience requirements.
- There seems to be a recent solution in economics literature for this issue (more literature review is definitely needed).
 - First, use an LLM to label a proper subsample. Humans then review LLM-generated labels; then this LLM-labeled subsample is splitted into the training and test set.
 - Second, train (fine-tune or apply) a deterministic model (often BERT-style models; [link to an introduction of BERT](#)) to minimize the prediction error in the training set.¹
 - Third, apply this deterministic model to the universe sample of texts and construct the desired variables.²

¹A good thing about BERT-style models is that they are deterministic (in contrast to LLM outputs which are often stochastic) and they are easy/cheap to scale.

²A preliminary step before this step is using the test set to measure the performance of different competing deterministic models.

Example 1 with this workflow

Algan et al. (2026): ([Link to paper](#))

We construct the scores in two stages. First, we draw a balanced random sample of 200,000 sentences, with 50,000 each from official-party tweets, citizen tweets, campaign speeches, and congressional floor speeches. We then ask GPT-4o-mini to assign a separate binary label for each of the nine dimensions based on the definitions and prompt in Appendix A.3.

Second, we use the GPT-labeled sentences to fine-tune distilbert-base-uncased as a multi-label classifier with nine independent sigmoid outputs. We fine-tune the model in-house using an 80/20 training-validation split and then apply it to approximately 142 million sentences across the observational datasets. For each sentence, the model produces a sigmoid score between zero and one for each emotional dimension.

We first convert each score to a binary indicator equal to one when the score exceeds 0.5. We test robustness to alternative thresholds. For analyses of the seven policy topics, we apply the topic filter at the document level: we retain every sentence in a document when at least one sentence in that document has a score above 0.5 for abortion, democracy, gender equality, gun control, immigration, tax and inequality, or trade. The topic scores are produced through an analogous two-stage procedure using a separate prompt (Appendix A.3). Climate tweets are selected by the phrases “climate change” and “global warming” and do not receive this additional policy-topic filter in the main time series. General tweets are also analyzed without

a policy-topic filter. Unless a figure or table states otherwise, descriptive emotion shares are sentence-level means across all retained sentences in the relevant dataset and period, with equal weight on each sentence. For topic-specific descriptive figures, a sentence may contribute to more than one topic when several topic scores exceed 0.5.

For the composition and time-series figures, we use the nine binary indicators to construct three mutually exclusive sentence categories. A sentence is *Angry* when its anger score exceeds 0.5. A sentence is *Other emotional* when it is not angry but either its holistic-emotionality score or at least one of the other seven focal-emotion scores exceeds 0.5. A sentence has *No detected emotion* when all nine scores are at or below 0.5. These categories exhaust the retained sentences and therefore sum to one within every topic, dataset, and period. They are derived through post-processing.

The detailed emotion-composition figures report the eight focal emotions as overlapping marginal shares. They also include *Other affect*, a derived category equal to one when the holistic-emotionality score exceeds 0.5 but none of the eight focal-emotion scores does. Because a sentence can express several focal emotions, the stacked marginal shares in these figures are not parts of a whole and need not sum to 100%.

Once trained, the classifier is a fixed standalone artifact that can be applied without querying the LLM again. Publishing the fitted classifier and its GPT-labeled training data therefore allows other researchers to reproduce the classification step using the same artifact. We make both available at <https://huggingface.co/thomasrenault/emotion>.

We assess the classifier against human annotations using a sample of 450 sentences, each independently classified by three human annotators (one author of the paper, one research assistant, and one external researcher). Each annotator is blind to the classifications of the other annotators and to the classification produced by the LLM. We find that LLM-generated annotations closely align with human judgments, consistent with recent evidence showing that LLMs can achieve high levels of reliability in classification tasks (Asirvatham et al., 2026; Niu et al., 2024). Appendix A.3.3 provides detailed validation results. As an additional face-validity check, we examine whether the language associated with each predicted emotion corresponds to intuitive and theoretically expected semantic patterns. Appendix A.3.3 presents word clouds highlighting the keywords most strongly associated with each emotion in the voter corpus. Across all nine dimensions, the most salient terms closely align with the conceptual meaning of the corresponding emotion.

Notes. LLM → fine-tuned DistilBERT as a deterministic multi-label clarification model → human validation

Example 2 with this workflow

Ash et al. (2025): ([Link to paper](#))

We first clean the dataset to focus on observations with interpretable text²¹, and then use a stratified sampling approach to select a sub-sample to annotate with the LLM. For each unique combination of two-digit NAICS industry and Glassdoor interview difficulty, we randomly select 500 observations. If there are fewer than that in the sample for a given combination, we use all such observations. This produces a dataset of 33,752 interview narratives.

Next, we use an aligned LLM to analyze the content of the interview descriptions in the sub-sample and annotate them across nine dimensions of interest. They are: colleague ability, colleague pleasantness, whether “they’d accept anybody,” direct assessment, compensation, hours and scheduling, workplace amenities, interview engagement, and realistic job preview. The LLM used is GPT-4o-mini and the prompt for each is available in Appendix D.

To ensure the quality of the annotations produced by the LLM, we hired two freelancers with experience in text annotation from the online freelancing platform Upwork. We provided the annotators with 100 interview descriptions, randomly sampled from this dataset of 33,752 narratives, and asked them the same prompts we used to train the LLM. Comparing the two annotators’ responses with those of the LLM, we observe an agreement score between human annotations and those of the LLM of over 75 percent (Appendix Table D1). The agreement rate of the LLM with the annotators is almost identical to the rate of agreement between the two annotators, indicating that we are close to the upper bound in terms of accuracy, given the subjective nature of these measurable concepts. We interpret the high agreement score as indicative of the LLM producing reliable interpretations for each prompt.

Last, for each prompt, we use the GPT-generated annotations to fine-tune a RoBERTa model on a binary classification task. We interpret labels of “Definitely Yes” and “Probably Yes” as the positive class, and the rest as the negative class. Each model is fine-tuned by optimizing hyper-parameters (i.e., learning rate, training batch size, and the number of training epochs). We then use the trained models to scale up the binary annotations for each prompt to all Glassdoor interview reports, not just those that received offers.

Notes. LLM with human validation → fine-tuned RoBERTa as a deterministic clarification model

Differences between our project and the previous 2 examples

Compared with the previous 2 examples, there are additional complexities for our project.

- The previous 2 examples try to measure something without a ground truth.
 - For example, “anger” in a sentence from example 1, “colleague pleasantness” from example 2.
 - However, for most of our variables constructed from job postings, they are involved with a yes or no question (e.g., does the job require a PhD degree).
- They only construct a small set (around 10) of binary variables using this workflow.
 - Even though their constructed variables are not necessarily well-defined, these generated binary variables are easy to use in future analysis.
 - However, in our project, we may construct more variables that are not necessarily binary.
- Actually, we don't have a very large sample size for the matched sample of new hires and job postings.
 - Recall from the last report, under the strict "time + state + job title" criterion, there are only 58,670 matched job postings, 21,044 matched new hires, and 1,320 companies with at least one matched new hire.

Outline

① General introduction: Variable construction from job postings

The goal

The measurement problem and one potential solution

My personal thoughts

② Variables: Education/experience requirements

③ Variables: Skill requirements

My personal thoughts

- There is some progress from economics and NLP/ML literature on a reasonable workflow to construct variables from job postings.
- Economics papers on job postings are still a little conservative about the variable construction method.
 - A recent paper on the usage of job postings (Audoly et al., 2026)³ still employs a keyword-matching method and involves little LLM or AI.
 - So we may not want to make a huge leap methodologically – otherwise it will be difficult to convince readers that our variable construction method is reasonable and robust.
- Otherwise, we may wish to capture something subtle in the job postings that would be hard to measure without AI.
 - Because we are trying to measure hiring practices within a narrow occupation-industry (instead of across occupations or industries), it could be very important to construct variables that have never been measured before in the literature.
 - For example, we may wish to capture the tradeoff between education and working experiences in some job postings.

³I will talk about this paper later because it is really important.

Relevant literature on the usage of job postings

- Modestino et al. (2020) (REStat) focuses on skill requirements that are similar to ours: They construct the share of postings requiring a bachelor's degree or greater and the share of postings requiring at least four years of experience at state-occupation-year level.⁴
- Deming and Kahn (2018) (JOLE) also focuses on skill requirements though as a different notion: There are 10 general skills, and a job posting requires 1 general skill (e.g., “social skill”) if it contains keywords like “communication,” “teamwork,” etc.
- Audoly et al. (2026) (R&R at AER, [link](#)) offers the most systematic workflow to construct variables (mainly wage and non-wage amenities) from job postings, and I highly recommend this paper.

► [Short summary](#)

I have tried my best to collect all economics papers that mainly use job postings, and a comprehensive literature review on this subject will be definitely helpful!

⁴The required vs. preferred degree requirements only show up in their robustness checks, and they don't consider the tradeoff between education and experiences, and they never explicitly say how the education/experience requirements are generated for each job posting.

Outline

- 1 General introduction: Variable construction from job postings
- 2 Variables: Education/experience requirements
 - General introduction: Education/experience requirements
 - Different formats for education/experience requirements
 - My current proposal to construction education/experience requirements
- 3 Variables: Skill requirements

What this section is about

- In this report, I will only focus on a subset of selectivity indicators: **Education and working experience requirements**.
 - Most job postings explicitly state these requirements, and they are direct proxies for hiring standard and selectivity level.
 - As we will see in some examples, the key complication comes from the fact that there are so many ways to format these requirements.
- Why do we need to construct these variables for our project?
 - Job advertisement is one of the few places where the firm's **ex ante screening standard** is directly observable.
 - Higher credential requirements and higher demands for relevant work experience are explicitly part of the **screening-intensive hiring strategy**.
 - In the end, we wish to distinguish this screening-intensive practices from **experimentation-oriented hiring strategies**.
- As I will show later, the main complexity comes from the fact that there are so many different formats for education/experience requirements in job postings.

Outline

① General introduction: Variable construction from job postings

② Variables: Education/experience requirements

General introduction: Education/experience requirements

Different formats for education/experience requirements

My current proposal to construction education/experience requirements

③ Variables: Skill requirements

Combined requirements (format 1) and minimum vs. preferred (format 2)

Example 1: Degree and experience requirements combined as a bundle (Job ID: 2149720339)

Requires 4+ years of laboratory experience with a Bachelors or better in Biology, Biotechnology, Chemistry, Chemical Engineering or related field.

Implication: Education and experience requirements are tightly linked.

Example 2: Minimum vs. preferred (Job ID: 2706164505)

Minimum of a bachelor's degree; advanced degree (Masters, PhD) in a science/health field, or equivalent, is desirable.

Implication: In our constructed variables, minimum and preferred requirements should be stored in separate variables.

Education-experience tradeoffs (format 3)

Example 3: Precise tradeoff (Job ID: 3659097343)

Masters with 1 year of experience or a BS with 3 or greater years of experience.

Implication: An advanced degree lowers required experience. In our constructed variables, a uniform minimum education/experience requirement variable can be misleading.

Example 4: Precise but more flexible tradeoffs (Job ID: 2708162985)

Doctorate degree; OR Master's degree AND 2 years; OR Bachelor's degree AND 4 years; OR Associate's degree AND 10 years; OR high school diploma/GED AND 12 years of Information Systems or Statistical Analysis experience.

Implication: This is more flexible tradeoff between education and working experience. We need to be able to capture such flexibility in our constructed variables.

Example 5: Mixed numeric forms (Job ID: 3868005283)

High School Diploma (or Equivalent) with 10 years; OR Associate's degree with eight years; OR Bachelor's degree with at least five years; OR Master's degree with more than three years; OR Ph.D. with 0–2 years of relevant experience.

Implication: There is again a tradeoff between education and experience. But the required experience are expressed in more complex ways (in ranges instead of exact numbers).

No-experience language (format 4)

Example 6: Ph.D. but no prior work experience (Job ID: 353284852400000000002)

Ph.D. in Immunology, Biology, Biochemistry, Pharmacology, Physiology, or related discipline. Postdoc or industry experience is not required. We specifically seek a recent Ph.D. graduate who wishes to forgo a postdoc and launch their career in industry. (The same posting though asks for “experience with” immunoassay platforms.)

Implications: (1) No working experience are specifically mentioned. (2) General working experience are different from task-specific experience.

Example 7: No experience conditional on education (Job ID: 3687992771)

*High School Diploma/Equivalent or Associate degree with two years GMP manufacturing/processing experience
OR Bachelor degree in Life Science or Engineering and no experience.*

Implication: Zero experience applies only conditional on education requirements.

Qualitative (ambiguous) experience requirements (format 5)

Example 8: No numeric duration (Job ID: 2912710843)

Ph.D. in Life Sciences discipline such as biology, neuroscience or biochemistry.

Initial post-doctoral or industry experience.

Hands-on experience in the development and validation of biomarker assays.

Experience in project and people management.

Example 9: Broad equivalency (Job ID: 3372358481)

Bachelor's Degree, preferably in Cell Biology and Molecular Biology, or equivalent relevant experience. Minimum 5 years of experience within the biologic, pharmaceutical or medical device industry. An equivalent combination of education and experience may substitute.

Implications: There are still experience requirements in the posting, though in a more ambiguous format. We should be able to distinguish from other postings where no experience requirements are explicitly mentioned.

Outline

- 1 General introduction: Variable construction from job postings
- 2 Variables: Education/experience requirements
 - General introduction: Education/experience requirements
 - Different formats for education/experience requirements
 - My current proposal to construction education/experience requirements
- 3 Variables: Skill requirements

Step 1: Construct different routes from job postings

Consider “Master’s + 1 year OR Bachelor’s + 3 years”:

Route	Education requirement	Experiences requirement
A	Master’s degree	At least 1 year
B	Bachelor’s degree	At least 3 years

- First, decompose the requirements into different routes. Each route is an acceptable or desirable education-experience combination that the job posting refers to.
- Each route documents two sub-vectors: education requirement and experience requirement.

Step 2: Vector representation for requirements in one route

Consider “PhD degree in biology or chemistry with 3+ years of experience is desirable.”

- We can construct a 3-dimension vector to represent the education requirement:
 - Degree: Bachelor, master, or PhD. (In the example, it is PhD.)
 - Field: Economics, CS, Biology, Chemistry, ... (In the example, it is biology or chemistry.)
 - Desirability: Preferred, or minimum (In the example, it is preferred.)
- We can construct a 3-dimension vector to represent the experience requirement:
 - Lower bound: 0, 1, 2... (In this example, it is 3.)
 - Upper bound: 4, 5, ..., or ∞ . (In this example, it is ∞).
 - Desirability: Preferred, or minimum (In the example, it is preferred.)

The education/experience requirements in the example is represented by a 6-dimension vector⁵:

(PhD, biology or chemistry, preferred, 3, ∞ , preferred)

⁵One fallback vector is needed to represent vague qualitative requirements – things like “An equivalent combination of education and experience may substitute.”

Step 3: Transform route-level vectors to posting-level variables

- The final step is to transform the route-level vectors to posting-level variables, which will be seen as characteristics at the job posting level.
- I didn't have time to think about all reasonable posting-level variables. Here are some examples that can be constructed:
 - Route structure (variables that can measure flexibility of the requirements⁶): `n_minimum_routes`, `n_preferred_conditions`, `edu_exp_tradeoff_any`, `any_qualitative_equivalency`.
 - Education requirements: `edu_minimum_any`, `edu_preferred_any`, `edu_unspecified`.
 - Experience requirements: `exp_required_any`, `exp_preferred_any`, `exp_explicit_zero_general_any`, `exp_unspecified`.

⁶If you know any paper that tries to capture this subtle tradeoff between education and experience, please let me know. I think most economics papers using job postings construct only 1 variable for education and 1 variable for experience.

Outline

- 1 General introduction: Variable construction from job postings
- 2 Variables: Education/experience requirements
- 3 Variables: Skill requirements**

Relevant papers and terminology

- Skill extraction (or other related concepts, entity recognition/extraction, skill classification) for job postings is an established task in the NLP literature:
 - A.-s. Gnehm et al. (2022), A.-S. Gnehm and Clematide (2020), 2024, Green et al. (2022), Krüger et al. (2025), Nguyen et al. (2024), Zhang, Jensen, et al. (2022), and Zhang, Van Der Goot, et al. (2023).
- The notion of skills in the NLP literature is very different from it in the economics literature. The main difference is the level of generality/aggregation.
 - Deming and Kahn (2018) focuses on 10 general skills⁷. A job posting requires a skill if it contains the corresponding keywords.
 - Take Zhang, Van Der Goot, et al. (2023) as another example in NLP literature.⁸
 - It trains a benchmark ML model to do 2 tasks: (1) **skill extraction** marks phrases in a job posting that express skill requirements, and (2) **skill classification** maps a marked phrase to an ESCO skill code.
 - For example, for a hypothetical sentence from a job posting “perform PCR, analyze sequencing data, and work in a team”, its model first extract 3 different phrases from this sentence (skill extraction) and then maps these 3 requirements to 3 distinct ESCO skill codes (skill classification).

⁷Cognitive, Social, Character, Writing, Customer service, Project management, People management, Financial, Computer (general), Software (specific).

⁸Its central reference is ESCO (European Skills, Competences, Qualifications and Occupations) fine-grained “skills/competences.” For example, for an animal therapist, “anatomy of animals” is essential and “use physiotherapy for treatment of animals” is optional.

ESCO and its relevance to our project

- ESCO works like a dictionary, and the purpose is describing, identifying and classifying professional occupations and skills relevant for the EU labour market and education and training area and systematically showing the relations between those occupations and skills ([official website](#)).
- From my understanding, it is useful for our project in two ways:
 - First, given an occupation (e.g., chemist), ESCO lists all essential and optional skills relevant for this occupation ([link](#)).
 - Second, it consists of a comprehensive list of skills (13,939 different skills). For example, “use chromatography software” is one skill in ESCO skill list ([link](#)).
- My current thoughts:
 - To extract skill requirements from job postings, it may be better if we start from a benchmark model in NLP literature (e.g., Zhang, Van Der Goot, et al., 2023) that maps each sentence in the job posting into a skill (or null) in ESCO skill list.
 - The good thing about this ESCO skill list is that we don’t need to compile our own list of skills, which requires a lot of domain knowledge about the Biopharma industry (and we don’t have it).
 - Instead of inventing every tool by ourselves, it may be better to apply existing ML models to our project to extract and classify skill requirements.

Outline

- ④ Appendix: Short summary of Audoly et al. (2026)
- ⑤ Appendix: References

The 47 binary variables constructed from job postings

[Back](#)

- The authors construct 47 binary variables in 10 categories using a dictionary-based method.
 - Compensation scheme (7 variables): Compensation level; competitive pay; collective-agreement pay; incentive pay scheme; hiring bonus; good overtime pay; any other mention of pay.
 - Financial attributes (3 variables): Pension scheme; insurance scheme; mortgage possibility.
 - Career opportunities (2 variables): Good career paths; on-the-job training.
 - Hours of work (3 variables): Full-time contract; part-time contract; full-time/part-time choice.
 - Convenient hours (3 variables): Possibility to work flexible hours; regular daytime work schedule; exemption from work-hour regulations.
 - Inconvenient hours (4 variables): Shift work; weekend/evening/night work; on-call employment; overtime work required.
 - Contract duration (3 variables): Permanent job; temporary job; fixed-term contract.
 - Workplace attributes (5 variables): Social environment; good colleagues; possibility for remote work; shared office space; inclusive work-life scheme.
 - Task-related attributes (7 variables): Interesting tasks; challenging tasks; variation in tasks; responsibility in the job; independence in performing tasks; leadership responsibility; work involving travel.
 - Minor perks (10 variables): Beautiful location; central location; company gym or sports team; parking on premises; company vehicle; any welfare scheme; company cabin; occupational health service; company canteen; flexible/extended holidays

The construction of a dictionary for each variable

- Humans read 1,200 randomly selected ads, listing whether a variable is mentioned in the ad and the expressions used to describe it.
- They train a Word2Vec continuous bag-of-words model on vacancy texts to find expressions occurring in similar contexts.
 - They search for similar phrases containing one, two, or three consecutive words.
 - They manually remove inappropriate suggestions.
 - For example, phrases meaning “flexible working hours” and “flexible work-time arrangements” belong to the flexible-hours dictionary. “Flexible job adaptation” is excluded because it need not imply control over working hours.
 - The final dictionary contains 1,772 unique expressions across the 47 attributes.
- They search the full vacancy text and the variable value is 1 if at least one expression in this variable’s dictionary occurs in the job ad.

Research question 1: What do employers advertise

What do employers advertise?

- About 60% of ads contain compensation-related information, but fewer than 10% give a salary number or bracket.
- Nearly all ads mention non-pay attributes.
- Industry, occupation, and location account for at most roughly 30-40% of variation.
- Adding establishment effects raises explanatory power to roughly 50-70%.

Research question 2: Do the ads convey reliable information?

- Question 2.1. Do advertised attributes predict independently measured employer quality?
 - Answer: Better employers generally advertise more attractive bundles. Ad content explains around 60% or more of cluster-level quality variation; meaningful explanatory power remains after controls. Non-pay content contributes more than pay content.
- Question 2.2. Do advertised attributes match the jobs workers actually obtain?
 - Answer: Advertised and realized attributes agree more than random matching would predict. Raw agreement ranges from approximately 76% to 98% across the reported attributes.
- Question 2.3. Do workers use the information when choosing between ads?
 - Experiment subjects choose between randomly paired real ads from the same sector and broad occupation.
 - Answer: Respondents value attributes such as flexible hours and remote work, dislike temporary employment and inconvenient hours, and tend to prefer ads from higher-quality employers.

Research question 3: How do advertised amenities contribute to inequality?

How do advertised amenities contribute to inequality?

- Advertised amenity utility is positively correlated with pay: it reinforces pay differences.
- Intrinsic amenity utility is negatively correlated with pay: it partly compensates for pay differences.
- Equalizing advertised amenities lowers employment-weighted mean pay by 2.3 log points and raises pay variance by about 13%.
- Equalizing intrinsic amenities raises mean pay by 3.8 log points and lowers pay variance by about 25%.

Their workflow to construct matched samples of new hires and job postings

Audoly et al. (2026) also matches new hires to job postings using Norwegian admin data.

- 1 First, they search for workers starting employment: (1) at the same establishment, (2) at the same four-digit occupation, and (3) within the six months following the posting date.
- 2 Second, they restrict links to establishment-occupation cells with only one hire in the following six months.
- 3 Third, they exclude ads posted on the same date in the same establishment-occupation cell.
- 4 Fourth, they remove hires of workers who previously had an employment spell with that employer. Once an employment spell is linked to a posting, they do not allow that same spell to be linked to another posting.

They have a very conservative match process, and the matched sample of job postings consist of around 11% of all postings in the full job postings dataset.

Differences and similarities between this paper and our project

[Back](#)

- This paper uses a pre-AI era method to construct variables systematically from job postings.
 - We can definitely construct the same set of variables using their compiled dictionaries.
 - However, they focus on the broad population while our focus is on inventors so their constructed variables may not be of particular interest to us. For our project, we wish to understand firms' hiring standard and selectivity level.
 - An important question is to what extent can we justify our variable construction methods.
- This paper also tries to match new hires to job postings for a subsample.
 - Their main focus is to show that the advertised contents are informative.
 - However, for our project, the combination of job postings and new hires' future career paths reveal a firm's hiring practices together; so this match is essential.

Outline

- 4 Appendix: Short summary of Audoly et al. (2026)
- 5 Appendix: References

References I

- Algan, Yann, Eva Davoine, Thomas Renault, and Stefanie Stantcheva (2026). “The Rise of Anger: Emotions and Policy Views”. *Working Paper*, pp. 1–186.
- Ash, Elliott, Soumitra Shukla, and Jason Sockin (2025). “Interviews”. *Working Papers*, pp. 1–85.
- Audoly, Richard, Manudeep Bhuller, and Tore Adam Reiremo (2026). “The Pay and Non-Pay Content of Job Ads”. *Working Paper*.
- Deming, David and Lisa B. Kahn (2018). “Skill Requirements across Firms and Labor Markets: Evidence from Job Postings for Professionals”. *Journal of Labor Economics* 36.S1, S337–S369.
- Gnehm, Ann-sophie, Eva Bühlmann, Helen Buchs, and Simon Clematide (2022). “Fine-Grained Extraction and Classification of Skill Requirements in German-Speaking Job Ads”. *Proceedings of the Fifth Workshop on Natural Language Processing and Computational Social Science (NLP+CSS)*. Abu Dhabi, UAE: Association for Computational Linguistics, pp. 14–24.
- Gnehm, Ann-Sophie and Simon Clematide (2020). “Text Zoning and Classification for Job Advertisements in German, French and English”. *Proceedings of the Fourth Workshop on Natural Language Processing and Computational Social Science*. Online: Association for Computational Linguistics, pp. 83–93.
- (2024). “Mapping Work Task Descriptions from German Job Ads on the O*NET Work Activities Ontology”. *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING 2024)*.

References II

- Green, Thomas, Diana Maynard, and Chenghua Lin (2022). “Development of a Benchmark Corpus to Support Entity Recognition in Job Descriptions”. *Proceedings of the Thirteenth Language Resources and Evaluation Conference*. European Language Resources Association, pp. 1201–1208.
- Krüger, Kai, Johanna Binnewitt, Kathrin Ehmann, Stefan Winnige, and Alan Akbik (2025). “Improving Online Job Advertisement Analysis via Compositional Entity Extraction”. *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*. Suzhou, China: Association for Computational Linguistics, pp. 27035–27053.
- Modestino, Alicia Sasser, Daniel Shoag, and Joshua Ballance (2020). “Upskilling: Do Employers Demand Greater Skill When Workers Are Plentiful?” *The Review of Economics and Statistics* 102.4, pp. 793–805.
- Nguyen, Khanh, Mike Zhang, Syrielle Montariol, and Antoine Bosselut (2024). “Rethinking Skill Extraction in the Job Market Domain Using Large Language Models”. *Proceedings of the First Workshop on Natural Language Processing for Human Resources (NLP4HR 2024)*. St. Julian’s, Malta: Association for Computational Linguistics, pp. 27–42.
- Zhang, Mike, Kristian Jensen, Sif Sonniks, and Barbara Plank (2022). “SkillSpan: Hard and Soft Skill Extraction from English Job Postings”. *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Seattle, United States: Association for Computational Linguistics, pp. 4962–4984.
- Zhang, Mike, Rob Van Der Goot, and Barbara Plank (2023). “ESCOXLM-R: Multilingual Taxonomy-driven Pre-training for the Job Market Domain”. *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*. Toronto, Canada: Association for Computational Linguistics, pp. 11871–11890.